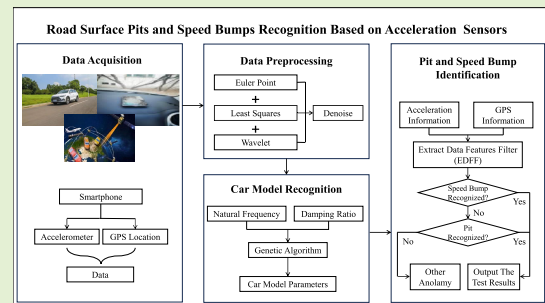


Road Surface Pits and Speed Bumps Recognition Based on Acceleration Sensor

Yunfei Yin^{ID}, Member, IEEE, Wanli Fu, Xianrong Ma^{ID}, Junpeng Yu, Xinkai Li, and Zejiao Dong

Abstract—Road potholes pose dangers to traffic safety, damaging vehicles, and increasing accident risks, especially during low visibility conditions. Traditional detection methods are costly, inefficient, and inaccurate. Leveraging advancements in smartphone sensor capabilities, this article presents a novel approach for road pothole and speed bump detection using smartphone acceleration and GPS sensors, along with extract data features filter (EDFF) technology. The method involves preprocessing acceleration data using Euler point, least squares, and wavelet techniques to enhance accuracy by reducing noise. The genetic algorithm is then employed to identify car model parameters. In addition, EDFF is established based on data features to differentiate between speed bumps and grooves and determine groove properties. Experimental results using real data from Harbin show pit recognition accuracy of over 75% and 100% accuracy for speed bumps. This demonstrates the proposal's effectiveness in detecting pits and speed bumps accurately, making it useful for practical engineering applications.

Index Terms—Acceleration sensor, extract data features filter (EDFF), pavement potholes, road detection, speed bumps.



I. INTRODUCTION

ONCE roads are used, they experience repetitive vehicle loads, leading to various surface issues such as potholes, cracks, and damage. Low-grade road surfaces face more severe problems due to the demanding driving loads they endure. Neglecting these issues poses a significant threat to the driver and pedestrian safety [1], potentially causing more accidents. However, maintaining road quality has been a challenging task for the road sector. Despite substantial investments, achieving optimal road conditions has remained elusive, hindering expectations. Detecting road surface diseases promptly

and effectively is crucial. This method should enable early identification of deteriorating road conditions for swift actions. Implementing such a method would safeguard drivers and pedestrians, reducing accidents and their consequences. Developing an efficient road disease detection method is essential for the well-being of all road users. With timely detection, appropriate measures can rectify road surface issues promptly, minimizing risks and enhancing road safety.

Initially, researchers mainly rely on manual testing methods to detect road surface diseases. For example, the level and ruler instruments are applied to detect flatness and road diseases [2]. Although these technologies can detect pavement conditions precisely, their detection effectiveness is limited, and it is unsuitable for wider surface inspections. In addition, the road disease information cannot be obtained and treated in time. With the advancement of sensor technology, a new approach to pavement detection has emerged, utilizing multiple sensors. Laser sensors have been employed to easily obtain surface flatness information [3]. Smartphone acceleration sensors have been proposed as a method to detect road surface flatness, improving detection effectiveness [4]. A study introduces a dynamic regularized least squares minimization method that utilizes vehicle acceleration to directly identify road roughness profile [5], demonstrating higher accuracy compared with the Kalman filter. A measurement method for characterizing pavement profile heights has been proposed using acceleration sen-

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Yunfei Yin, Xianrong Ma, Junpeng Yu, Xinkai Li, and Zejiao Dong are with the School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin 150090, China (e-mail: yunfeiyin@hit.edu.cn; maxianrong@hit.edu.cn; hitdzj@hit.edu.cn).

Wanli Fu is with China State Construction Engineering (Hong Kong) Ltd., Hong Kong, China (e-mail: fuwanli124@cohl.com).

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sors and pavement contour optimization technology [6]. Novel methods for pavement pit detection, such as using microphones, cameras, and laser imaging, have been developed for more efficient pit inspection [7], [8], [9], [10]. A methodology utilizing ground penetrating radar (GPR) for detecting moisture-related stripping damage in asphalt pavements has been developed [11], although it is influenced by ambient conditions and exhibits high cost and low detection accuracy.

With the advancement of global road construction, there is a growing need for pavement inspection information. Relying solely on sensors for data acquisition no longer suffices. Researchers have turned to pavement modeling to predict diseases and gain comprehensive information. A bituminous pavement model combines nonlinear damage analysis and fracture mechanics to predict localized behavior [12]. A methodology has been proposed to back-calculate damage density in asphalt pavements, enabling cost-effective maintenance [13]. A stochastic model based on empirical-mechanistic methods anticipates flexible pavement cracking [14]. However, disease prediction based on modeling has limitations. Recently, image processing techniques have been explored for detecting potholes and cracks [15], [16], [17], but the method relies on collecting road surface images, affected by environmental factors, and accuracy needs improvement.

The proposal of a smartphone-based method for road surface pits and speed bump recognition is motivated by the wide usage of smartphones in various fields [18], [19], [20]. For instance, the RoadSense Android application utilizes smartphone sensors and a decision tree classifier to predict road quality with high accuracy [21]. Another study employs smartphone sensor data and deep neural networks for classifying road surface conditions, highlighting its potential for large-scale implementation [22]. In addition, a method utilizing accelerometer data and fast Fourier transforms is proposed for estimating pavement unevenness [23]. Based on these experiences, the present study introduces a method using smartphone acceleration and GPS sensors to detect road potholes and speed bumps. Data collection involves installing four smartphones in a car and preprocessing the acceleration data to remove noise. The proposed method employs the extract data features filter (EDFF) technology, utilizing the differences in acceleration peaks and troughs to accurately identify pits and speed bumps. Compared with other detection technologies, the proposed method offers high detection effectiveness and helps reduce pavement inspection costs.

The remainder of this article is organized as follows. Section II gives a theoretical analysis of data preprocessing, car model parameter inversion, and pit and speed bump identification system. Section III uses simulation software to verify the theory of pit and speed bump identification. Section IV conducts the calculation of vehicle model parameters and the identification of pits and speed bumps through outdoor experiments. Section V provides the conclusion of this article.

II. THEORETICAL RESEARCH

Fig. 1 shows the schematic flowchart of the proposed road surface pits and speed bump recognition based on a smartphone platform. First, collect the acceleration and GPS

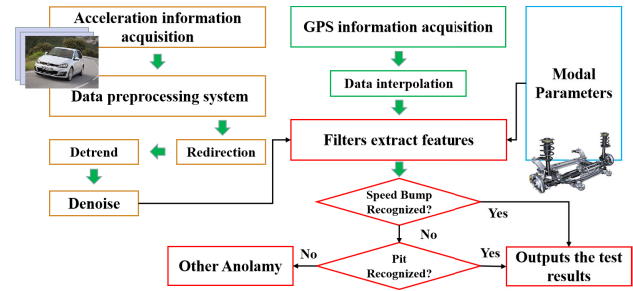


Fig. 1. Schematic flowchart of the proposed road surface pits and speed bumps recognition.

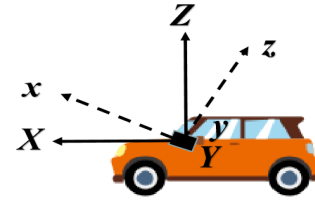


Fig. 2. Deviations of the triaxial direction.

data using the mobile phone; then, the data are preprocessed through three stages, i.e., data redirection, detrending items, and data noise reduction. Second, the genetic algorithm is exploited to identify the physical parameters of the vehicle, obtaining accurate road excitation. Finally, the EDFF method is proposed to identify the road pits and speed bumps.

A. Data Preprocessing

The gathered data from the mobile phone acceleration sensor in the automobile have noticeable noise due to the influence of body self-vibration. In addition, with the continuous growth of the test distance, there is a baseline drift. Therefore, the collected acceleration data need to be preprocessed. The data preprocessing include three stages, i.e., data redirection, detrending items, and data noise reduction.

1) *Data Redirection*: When the sensor is mounted on the vehicle for acceleration data collection, there exist variances among x , y , and z triaxial directions of the sensor and triaxial directions of the car, as illustrated in Fig. 2. When the deviation between the sensor and the automobile is large, it can result in a large difference in the measurement results from the real situation.

This article implements the reorientation of the data by utilizing the Euler angles method, which has three independent parameters that allow the orientation of any object in space through a series of fundamental rotations [24], [25]. During the redirection process, the angle of rotation around the x -axis is set to α (rolling angle), the angle of rotation around the y -axis is β (pitch angle), and the angle of rotation around the z -axis is γ (yaw angle).

When the vehicle is stationary, the acceleration of z -axis is gravity acceleration, while the accelerations of both x -axis and y -axis are 0, i.e., as follows:

$$\begin{aligned} a_x &= 0 \text{ m/s}^2 \\ a_y &= 0 \text{ m/s}^2 \\ a_z &= 9.8 \text{ m/s}^2 = 1 \text{ g}. \end{aligned} \quad (1)$$

Setting the raw acceleration as a'_x , a'_y , and a'_z , there are two Euler angles as follows:

$$\alpha = \tan^{-1}\left(\frac{a'_y}{a'_z}\right)$$

$$\beta = \tan^{-1}\left(\frac{-a'_x}{\sqrt{(a'_y)^2 + (a'_z)^2}}\right). \quad (2)$$

The yaw angle γ can be reoriented to the standard X - and Y -axes by the original data axes x - and y -axes. The reoriented acceleration along the three axes can be estimated by

$$\begin{aligned} a_x &= a'_x \cos(\beta) + a'_y \sin(\beta) \sin(\alpha) + a'_z \cos(\alpha) \sin(\beta) \\ a_y &= a'_y \cos(\alpha) + a'_z \sin(\alpha) \\ a_z &= -a'_x \sin(\beta) + a'_y \cos(\beta) \sin(\alpha) + a'_z \cos(\alpha) \cos(\beta). \end{aligned} \quad (3)$$

In addition, the range of the rolling angle α is defined as $[-\pi, \pi]$, and the range of the pitch angle β is defined as $[-\pi/2, \pi/2]$.

2) Detrending Items: When using a mobile phone for acceleration data acquisition, the data produce the trend item due to factors, such as temperature and slope. This may cause baseline drift which in turn affects the accuracy of the calculation. Thus, this step is to eliminate the trend item of the data to increase data accuracy.

When removing trend items, the data can be processed using least squares. The trend item of acceleration data collected by micro-electromechanical systems sensor deviating from the time baseline generally presents a linear trend. Therefore, the least squares method is used to eliminate the trend term based on the linear principle.

3) Data Noise Reduction: In this article, the wavelet threshold denoising method is utilized to eliminate acceleration signal data noise. The wavelet transform is a technique for distinguishing between real and noisy signals. The wavelet coefficients obtained after wavelet transformation are generally high because the energy of actual signals is comparatively concentrated. However, the ability of noisy signal is relatively distributed, and the wavelet coefficients obtained after wavelet transformation are often tiny [26], [27], [28]. According to this principle, a wavelet threshold noise reduction method is proposed. This method of data processing has fewer constraints, greater noise reduction and is more universal in real signals.

When the wavelet threshold noise reduction is performed in this article, the wavelet base function selected is the Symlet wavelet base, which is more suitable for processing acceleration data. The soft threshold value method for data processing selects the threshold value, which is a smoother data processing method. The data are processed by selecting the three-layer wavelet decomposition method after comparing various decomposition layers. Fig. 3 presents the data preprocessing of a set of acceleration data gathered from the mobile phone.

B. Car Model Recognition

When using the vibration response of the vehicle for road excitation detection, vehicle model parameters can impact the

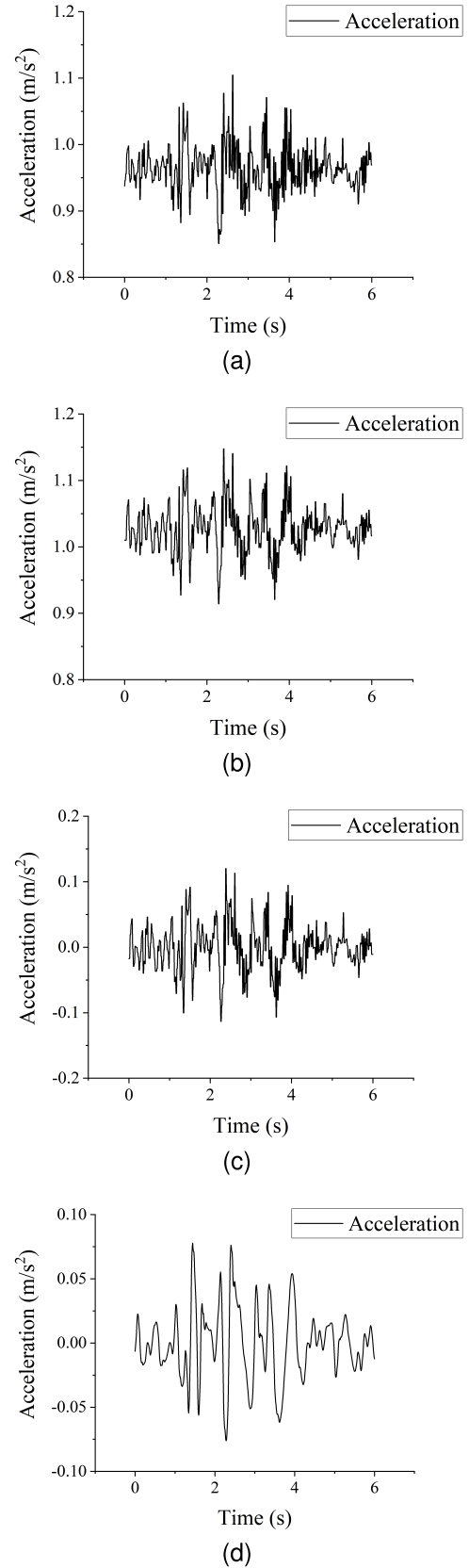


Fig. 3. Acceleration data preprocessing. (a) Raw acceleration data. (b) Data redirection. (c) Data detrending. (d) Data noise reduction.

data obtained in the experiment. Thus, to obtain more accurate road excitation information, the physical parameters of the vehicle need to be identified. This challenge can be handled

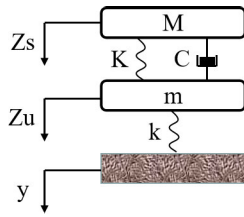


Fig. 4. Quarter car suspension model.

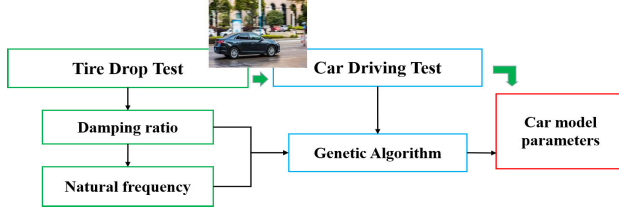


Fig. 5. Technology roadmap.

by converting the physical properties of the vehicle into model parameters and then inverting the parameters of the vehicle model. However, an obvious disadvantage of this method is that the excitation of the pavement is difficult to determine which is usually simulated with expensive equipment.

In this article, a two-sensor joint test approach is presented, which collects acceleration data from the vehicle body part as well as the wheel part. The wheel acceleration is regarded as the incentive to give the suspension, and the body acceleration is the most excitatory response, as shown in Fig. 4, and its dynamic equation is expressed as follows:

$$\begin{aligned}\ddot{Z}_s &= \frac{-K}{M}Z_s - \frac{C}{M}\dot{Z}_s + \frac{K}{M}Z_u + \frac{C}{M}\dot{Z}_u \\ \ddot{Z}_u &= \frac{K}{m}Z_s + \frac{C}{m}\dot{Z}_s - \frac{K+K_t}{M}Z_u - \frac{C}{m}\dot{Z}_u + \frac{K_t}{m}y\end{aligned}\quad (4)$$

where M is the mass of the sprung part, m is the mass of the unsprung part, K is the spring stiffness coefficient of the sprung part, k is the spring stiffness coefficient of the unsprung part, C is the damping coefficient of the sprung part, $\ddot{Z}_s$ is the acceleration of the sprung part, $\ddot{Z}_u$ is the acceleration of the unsprung part, and y is the pavement profile.

Finally, the car model parameters are obtained by establishing a genetic algorithm, as shown in Fig. 5. In general, the car model parameter recognition is divided into two parts: model parameter identification and physical parameter recognition.

1) Model Parameters:

a) *Natural frequencies*: The natural frequency is the displacement of the object changes with time according to the law of sine/cosine when doing free vibration. The frequency of the vibration is independent on the initial conditions, only related to the inherent characteristics of the system. The natural frequency of the vehicle suspension system is mainly divided into two states, namely, the undamped natural frequency and damped natural frequency, and there is a certain conversion relationship between them

$$f_0 = f_1 \sqrt{1 - (\zeta)^2} \quad (5)$$

where f_0 is the natural frequency of the damped state, f_1 is the natural frequency of the undamped state, and ζ is the damping ratio.

When the on-spring or under-spring part does not move, the frequency at which the other part vibrates is known as offset frequency. The offset frequency can be used to verify the dynamic model, estimate the on-spring mass and the under-spring mass, and analyze the high-frequency and low-frequency resonance characteristics of car. The offset frequency is very similar to the natural frequency of the two degree-of-freedom system, according to the existing analysis, it can be substituted by the offset frequency to produce an undamped natural frequency.

Therefore, when the system experiences excessive free vibration, the quarter car model can be used to determine the relationship between the spring-on natural frequency f_1 and the subspring natural frequency f_2 and the physical parameters of the vehicle as follows:

$$\begin{aligned}f_1 &= \frac{1}{2\pi} \sqrt{\frac{K}{M}} \\ f_2 &= \frac{1}{2\pi} \sqrt{\frac{K+K_t}{M}}.\end{aligned}\quad (6)$$

b) *Damping ratio*: The damping ratio refers to the ratio of the damping coefficient to the critical damping coefficient, which reflects the attenuation form of vibration of the structure after it has been activated. It is a key term in structural dynamics in the disciplines of equipment and automobiles. The damping ratio is determined in this work using the logarithmic recognition approach. The half-cycle decay rate of the acceleration curve $\tau = A_1/A_2$ of the body when the car falls is calculated, where A_1 is the difference between the second peak and the third peak, and A_2 is the difference between the third peak

$$\begin{aligned}C_r &= 2M\omega_n = 2\sqrt{KM} \\ \zeta &= \frac{C}{C_r} = \frac{C}{2M\omega_n} \\ \zeta &= \frac{1}{\sqrt{1 + \frac{\pi^2}{\ln^2 \tau}}}\end{aligned}\quad (7)$$

where C_r is the critical damping coefficient, C is the damping coefficient, and ζ is the damping ratio.

2) *Vehicle Physical Parameters Recognitions*: Genetic algorithm is a computational model that simulates the natural selection and genetic mechanisms of Darwin's theory of biological evolution and is a method of searching for optimal solutions by simulating the process of natural evolution. Through mathematical means and computer simulation operations, the algorithm converts the problem-solving process into a process similar to the intersection and mutation of chromosomal genes in biological evolution.

For ease of analysis physical parameter information of the vehicle, the traditional quarter-model can be simplified as follows:

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-K}{M} & \frac{C}{M} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} [\ddot{Z}_u] \quad (8)$$

$$[\ddot{Z}_s] = \begin{bmatrix} \frac{-K}{M} & \frac{-C}{M} \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}. \quad (9)$$

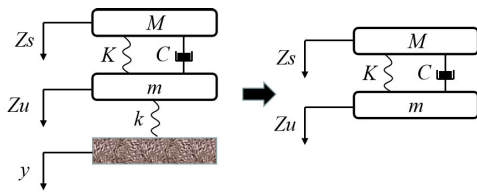


Fig. 6. Simplify the model of the car.

The simplified model removes the road excitation part, and the structure is composed of a structure that provides excitation with the under-spring mass and responds to the on-spring mass as shown in Fig. 6.

In this article, a genetic algorithm model is developed by placing the sensor on the front and right front wheels of the co-pilot to obtain the true values of the on-reed and subreed acceleration. After that, the acceleration under the spring is used as the excitation to reverse the acceleration on the spring and compare calculated values to its true value to minimize the error. Subsequently, the parameter values of the vehicle K/M and C/M are obtained, and the model parameters, such as natural frequency and damping ratio, are used to obtain the parameter values of m/M and k/M . As demonstrated in (8), the relationship between road excitation and body reaction is based on the ratio of various physical factors; therefore, the ratio of physical parameters determined using genetic algorithms fits the body response calculation requirements.

C. Pit and Speed Bump Identification

When the car is driving on the road, the body vibration signal is composed of intermittent vibration signals, such as pits and speed bumps on the one hand, and continuous vibration signals caused by the self-vibration and flatness of the body. The goal of this identification method is to extract and identify these unique data in order to detect pits and speed bumps.

1) **Signal Rejection:** When the car is driving on the road, continuous fluctuation data will appear due to the influence of body vibration and road surface smoothness. This vibration signal can be filtered by setting a threshold value, and all are classified as 0. However, as the speed of the car increases, the vibration induced by the smoothness of the road surface intensifies, necessitating the adjustment of the specified threshold value to the vehicle speed to avoid data omission or excessive filtering.

2) **Identification of Potholes and Speed Bumps:** Potholes and speed bumps have distinct structures; thus, the characteristics of the acceleration signal generated by the car after driving through potholes and speed bumps are diverse as well. The acceleration signal first appears as a trough, then generates a peak, since the pothole is a type of road depression. However, the speed bump is a type of road bump; therefore, the acceleration signal will first appear as a greater peak, followed by a trough.

3) **Pit Attribute Recognition:** When the car goes through pits of varying depths, the body response generated by shallower pits is smaller, while the body response induced by deeper pits is larger. According to this law, thresholds can be set for depth detection of pits. The determination of the pit width

TABLE I
POTHOLE AND SPEED BUMP DETECTION THEORY

```

for each point (x,y)
   $N_{pit}=0, N_s b=0$ //initialize
  for each input data
    Identify peaks threshold  $I_p$ ;
    Identify troughs threshold  $I_t$ ;
    Compute  $n$ ;
    Identify peaks  $(x_1, y_1)$ ;
    Identify peaks  $(x_2, y_2)$ 
  for each point (x,y)
    if  $y > I_p$ 
       $a_1 = x$ 
    else if  $y < I_t$ 
       $a_2 = x$ 
    end for
    if  $n > a_1 - a_2 > 0$ 
       $N_p = N_p + 1$ 
    else if  $n > a_2 - a_1 > 0$ 
       $N_s = N_s + 1$ 
    end for

```

TABLE II
CAR MODEL PARAMETERS

Parameter	Spring-on mass	Damping	Spring rate	Tire mass	Tire rate
Target vehicle	1200 Kg	10 N/(m/s)	180 N/mm	87 Kg	1962 N/mm

can be accomplished by integrating the starting point of the characteristic signal trough and the starting point of the peak in the time domain as the car goes through pits of various widths. When a car goes through potholes at varying speeds, the signals of the shallower potholes become weaker over time as the vehicle speed increases, while the signals of the deeper potholes become more visible. Therefore, when the car travels at different speeds, the set threshold needs to be an initial adjustment.

The above three parts together constitute the theory of pit groove and speed bump identification, which is the prototype of the EDFD technology. The specific detection algorithm is shown in Table I.

III. SIMULATION RESULTS

In this section, a simulation result is provided to verify the feasibility of the proposed pit and speed bump identification approach. The simulation is divided into three steps. First, use simulation software Carsim to simulate the acceleration data of the car driving through the pits and speed bumps. Then, analyze the influence of vehicle speed and groove attributes on acceleration data, and finally, use the proposed algorithm to identify pits and speed bumps and pits.

A. Design Conditions

Vehicle parameters are based on C-type sedan of Carsim, and the parameters, such as body suspension mass, spring progress coefficient, and tire quality, are set as shown in Table II.

According to the definition of potholes in highway regulations, road potholes refer to potholes with a depth of not less than 2 cm, which is prone to occur frequently on asphalt pavements common disease. According to the common

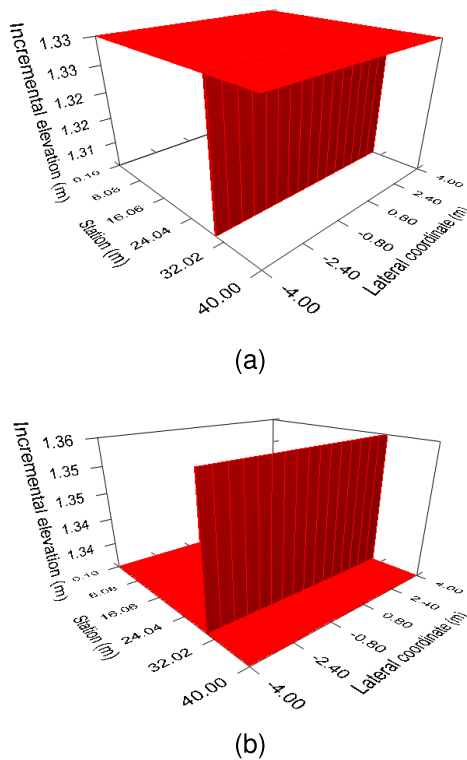


Fig. 7. Carsim pits and speed bumps model. (a) Pit. (b) Speed bump.

TABLE III
DESIGN OF PIT CONDITIONS

Speed	2~3 cm	3~4 cm	4~5 cm	5 cm or more
20 km/h	C-P1	C-P5	C-P9	C-P13
30 km/h	C-P2	C-P6	C-P10	C-P14
40 km/h	C-P3	C-P7	C-P11	C-P15
50 km/h	C-P4	C-P8	C-P12	C-P16

TABLE IV
DESIGN OF SPEED BUMP CONDITIONS

Speed	3 cm	4 cm	5 cm
10 km/h	C-S1	C-S6	C-S11
20 km/h	C-S2	C-S7	C-S12
30 km/h	C-S3	C-S8	C-S13
40 km/h	C-S4	C-S9	C-S14
50 km/h	C-S5	C-S10	C-S15

potholes and driving conditions on the road, the specific working condition design is shown in Fig. 7. The design of pit conditions and speed bump conditions are presented in Tables III and IV, respectively, where C-P stands for the condition of the pit and C-S stands for the condition of the speed bump.

B. Data Analysis

Simulate each working condition to obtain Z-axis acceleration data and vehicle speed. In the simulated environment, the road surface flatness and the natural vibration of the vehicle will not have a greater impact on the data. Therefore, the data preprocessing part can be omitted.

As shown in Figs. 8 and 9, when the car is driving through potholes and speed bumps, there are obvious peaks and valleys in the acceleration curve. However, because of different structural forms, the acceleration curve corresponding to the speed

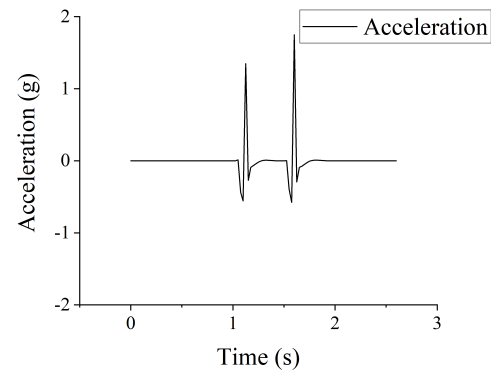


Fig. 8. Acceleration data of pits road conditions.

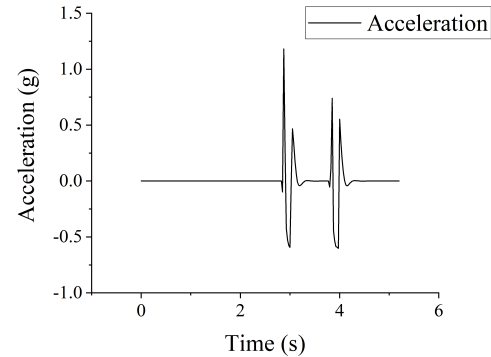


Fig. 9. Acceleration data of speed bumps road conditions.

TABLE V
DATA FOR ONE-SIDED WHEELS DRIVING THROUGH POTHOLES

Speed	2~3 cm		3~4 cm		4~5 cm		5 cm or more	
	P	T	P	T	P	T	P	T
20 km/h	0.65	-0.26	1.26	-0.26	1.01	-0.28	0.79	-0.29
30 km/h	0.53	-0.26	0.53	-0.28	0.43	-0.28	0.27	-0.29
40 km/h	0.42	-0.27	0.33	-0.24	0.25	-0.28	0.24	-0.29
50 km/h	0.49	-0.25	0.49	-0.25	0.34	-0.25	0.27	0.24

TABLE VI
DATA FOR DOUBLE-SIDED WHEELS DRIVING THROUGH POTHOLES

Speed	2~3 cm		3~4 cm		4~5 cm		5 cm or more	
	P	T	P	T	P	T	P	T
20 km/h	1.40	-0.51	2.22	-0.56	1.96	-0.56	1.99	-0.59
30 km/h	1.04	-0.52	0.98	-0.56	0.82	-0.56	0.49	-0.56
40 km/h	0.84	-0.55	0.63	-0.55	0.43	-0.54	0.46	-0.54
50 km/h	0.93	-0.49	0.79	-0.48	0.63	-0.46	0.48	-0.45

bump first shows an obvious peak, then a valley. However, the acceleration change caused by potholes is the opposite, which is the main feature of distinguishing potholes and speed bumps, and the main basis for road anomaly identification. After simulating various working conditions, the peaks and valleys of the acceleration curve are obtained as shown in Tables V–VII, and the width of the pits is 20–30 cm, where *P* refers to the peak of the wave and *T* refers to the troughs of the wave.

Subsequently, the pit groove and speed bump identification are carried out according to the characteristics of the acceleration data change. Distinguish the pit properties under different speed conditions by setting filters. Finally, the above logic is encoded, and the program development is preliminarily completed.

TABLE VII
PEAK INFORMATION OF SPEED BUMPS

Speed	3~4 cm		4~5 cm		5 cm or more	
	P	T	P	T	P	T
10 km/h	1.18	-0.60	3.35	-0.62	3.62	-0.60
20 km/h	1.50	-0.60	3.57	-0.61	4.77	-0.62
30 km/h	1.88	-0.61	3.84	-0.63	5.39	-0.68
40 km/h	2.82	-0.66	4.06	-0.66	6.56	-0.71
50 km/h	3.45	-0.67	4.75	-0.67	7.37	-0.69

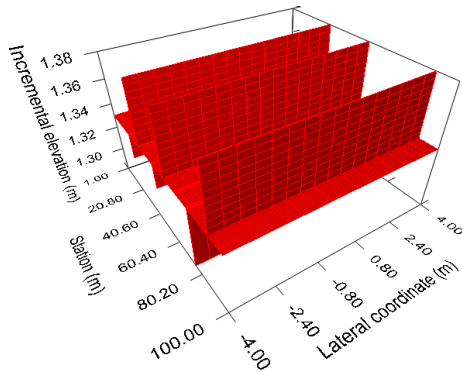


Fig. 10. Inspection of working conditions.

TABLE VIII
GROOVE AND SPEED BUMP TRUE PROPERTIES

Construction number	Structural type	Pit depth	Pit width
1	pit	2.3 cm	25 cm
2	speed bump	/	/
3	pit	4.7 cm	35 cm
4	speed bump	/	/
5	pit	6.5 cm	33 cm
6	pit	5.8 cm	37 cm
7	speed bump	/	/

TABLE IX
TEST RESULTS OF PITS AND SPEED BUMPS

Construction number	Structural type	Pit depth	Pit width	Pass time
1	pit	2~3 cm	20 cm	2.500 s
2	speed bump	/	/	/
3	pit	4~5 cm	30 cm	5.825 s
4	speed bump	/	/	/
5	pit	5 cm or more	30 cm	9.425 s
6	pit	5 cm or more	30 cm	8.200 s
7	speed bump	/	/	/

The design conditions are redesigned as shown in Fig. 10 after the code is produced to identify the pits and speed bumps to evaluate the practicality of the proposed algorithm.

The data obtained by running the program is processed to determine the pits and speed bumps contained in the road section, the pit grooves and speed belt settings are shown in Table VIII, and the test results are as shown in Table IX.

IV. EXPERIMENT

A. Vehicle Model Recognition Experiment

The car model recognition experiment mainly consists of two parts: the tire drop experiment and the car driving experiment. Among them, the tire drop experiment is used to identify the model parameters, and the car driving experiment



Fig. 11. Roll-down method experiments and sensor placement.

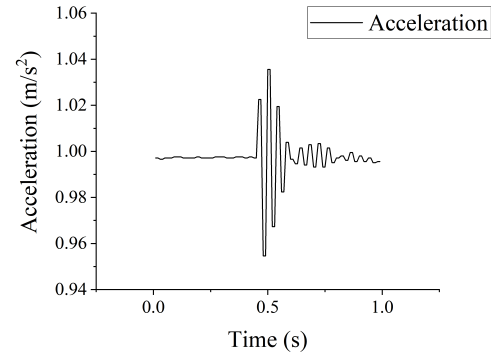


Fig. 12. Car drop acceleration.

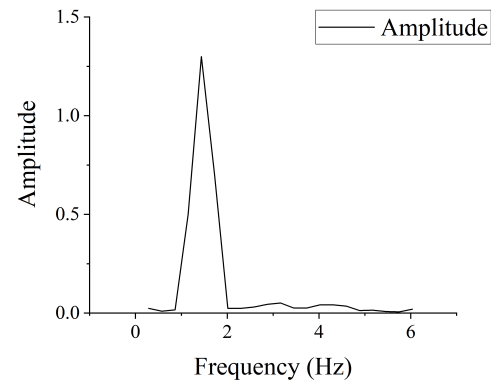


Fig. 13. Acceleration mean square spectrum.

is used to establish a genetic algorithm to identify the physical parameters of the car.

1) *Tire Drop Experiment*: To obtain the natural frequency, we designed a tire drop experiment shown in the Fig. 11. First, the car is driven to a block with a height of 90 mm. Second, after parking and hanging the neutral gear, the engine is turned off, and then, the car wheel is pushed down from the bump, trying to ensure that the left and right wheels land at the same time. In meanwhile, fix the MEMS sensor in the center of the tire and the front of the co-pilot and collect the acceleration data when the car falls. Then, the collected data are self-spectrally processed to obtain the natural frequency data shown in Figs. 12 and 13. The damping ratio can also be calculated from the vibration data of difference among the second peak, the third peak, and the fourth peak obtained from the tire drop. Then, use the equation half-cycle decay rate of the acceleration curve $\tau = A_1/A_2$ of the body and the equation for calculating the damping ratio (7) implement damping calculations. The calculation results of the parameters are obtained by integrating the experimental data of each working condition, as shown in Table X.

2) *Car Driving Experiment*: The car driving experiment is an experiment that uses sensors fixed to the interior and tires of the car to collect the acceleration data of the car on the

TABLE X
NATURAL FREQUENCY AND DAMPING RATIO ACQUISITION

The test number	Spring-on natural frequency	Damping ratio	Sub-spring natural frequency
Test1	1.179	0.294	13.1
Test2	1.268	0.278	10.9
Test3	1.112	0.268	11.6
Test4	1.296	0.272	12.1
Test5	1.252	0.288	11.3
Standard Deviation	0.075	0.011	0.848
Coefficient	0.061	0.039	0.072
Mean	1.221	0.280	11.8

TABLE XI
VEHICLE MODEL PHYSICAL PARAMETERS

Speed	The test number	K/M	C/M	m/M	k/M
20km/h	Test1	59.841	3.327	0.144	654.041
	Test2	58.626	3.241	0.139	655.256
	Test3	58.354	3.270	0.143	655.528
	Test4	57.732	3.152	0.134	656.150
	Test5	56.188	3.223	0.144	657.694
30km/h	Test6	57.931	3.475	0.162	655.951
	Test7	62.752	3.824	0.182	651.130
	Test8	63.630	3.351	0.138	650.252
	Test9	61.221	3.661	0.171	652.661
	Test10	59.374	3.185	0.133	654.508
40km/h	Test11	60.121	3.341	0.144	653.761
	Test12	63.425	3.513	0.152	650.457
	Test13	58.747	3.252	0.140	655.135
	Test14	59.721	3.417	0.152	654.161
	Test15	62.733	2.934	0.107	651.149
	Standard Deviation	2.192	0.209	0.016	2.192
	Coefficient	0.036	0.062	0.116	0.003
	mean	60.034	3.336	0.145	653.848

spring and under the spring during driving. The sensor is laid out in the same way as the tire drop experiment shown in Fig. 11. This experiment provides a data basis for the construction of genetic algorithms. Use the acceleration data at the tire as the excitation, and use (8) for inverse calculation of body acceleration data. Comparing the calculated values with acceleration data of the body collected by the sensor, the car model parameter values are obtained by continuously adjusting the parameters. The values of various parameters obtained based on genetic algorithms and automobile driving experiments are shown in Table XI.

The final obtained car model parameters are $K/M = 60.034$, $C/M = 3.336$, $m/M = 0.145$, and $k/M = 653.848$.

B. Pit and Speed Bump Identification Experiment

The proposed pit and speed bump recognition approach based on smartphones is used to detect the state of road pits and speed bumps in the road sections of Harbin to verify its feasibility. A number of typical pits and grooves in Harbin are initially selected to collect vehicle vibration data, and the identification system is developed according to the rules and regulations.

1) *Data Acquisition and Preprocessing*: Select the road sections that meet the requirements to obtain the actual pits and

TABLE XII
PIT TEST CONDITIONS

Condition1 10~20 km/h	Condition2 20~30 km/h	Condition3 30~40 km/h
test1	test4	test7
test2	test5	test8
test3	test6	test9



Fig. 14. In-vehicle information collection equipment.

TABLE XIII
SENSOR ARRANGEMENT

Device	Number	Location
Acceleration acquisition equipment	3 units	Above the front tire and at the center
GPS data collection equipment	1 units	Driver's front

TABLE XIV
SPEED BUMP INFORMATION

Speed bump	Right front1		Right front2		Right front3	
	P	T	P	T	P	T
10 km/h	4.45	-3.92	4.33	-3.33	3.29	-3.12
20 km/h	5.41	-3.54	7.10	-4.72	4.85	-5.33
30 km/h	7.71	-8.13	10.37	-12.28	7.44	-7.22
	Left front1		Left front2		Left front3	
	P	T	P	T	P	T
10 km/h	3.82	-4.20	4.03	-3.88	3.52	-3.35
20 km/h	5.16	-3.00	5.85	-6.22	5.56	-5.64
30 km/h	7.41	-8.42	9.59	-11.12	7.31	-7.21
	Middle1		Middle2		Middle3	
	P	T	P	T	P	T
10 km/h	3.03	-2.85	3.02	-2.92	2.66	-2.69
20 km/h	3.96	-2.64	5.46	-4.13	4.81	-4.01
30 km/h	6.08	-6.97	7.08	-7.55	6.03	-5.21

speed bump vibration data. The driving speed is limited to 10–40 km/h, and each pit is driven three times at each speed as a parallel experiment to ensure data quality. The specific working conditions are designed as shown in Table XII.

During data collection, to obtain information such as the attribute and location of the pit, it is necessary to obtain acceleration and GPS data of multiple positions inside the car. Considering the possibility that both the left and right wheels of the vehicle pass through the pit as shown in Fig. 14 and the sensor arrangement is given in Table XIII.

The Z-axis acceleration information graph and wave peak and trough information collected in the experiment are as shown in Tables XIV and XV, and the preprocessed acceleration data is shown in Figs. 15 and 16.

2) *Identification Algorithm*: Analyze the obtained data of various working conditions, and realize the identification of speed bumps and pits by designing filters. According to the theory of pit and speed bump identification in Section II, the pit identification filters are mainly composed of five parts.

a) *Data fusion*: Because each side of the tire may pass through the pit during the driving process of the vehicle, it is necessary to combine the data of the left sensor and the right sensor as shown in Fig. 17. After retaining the data points

TABLE XV
PITS INFORMATION (RIGHT 2.5 cm)

Speed bump	Right front1		Right front2		Right front3	
	P	T	P	T	P	T
10 km/h	1.22	-1.15	1.13	-0.94	1.39	-1.54
20 km/h	1.80	-1.91	1.97	-2.17	1.93	-2.30
30 km/h	2.09	-2.68	1.92	-2.35	2.07	-2.97
	Left front1		Left front2		Left front3	
	P	T	P	T	P	T
10 km/h	0.31	-0.75	0.36	-0.44	0.46	-0.69
20 km/h	0.82	-1.48	0.88	-1.04	0.93	-1.08
30 km/h	1.16	-0.80	1.53	-0.72	1.53	-0.90
	Middle1		Middle2		Middle3	
	P	T	P	T	P	T
10 km/h	0.45	-1.01	0.41	-0.35	0.45	-0.60
20 km/h	1.39	-0.79	0.61	-1.03	0.77	-1.44
30 km/h	1.16	-1.17	1.36	-1.27	1.29	-1.52

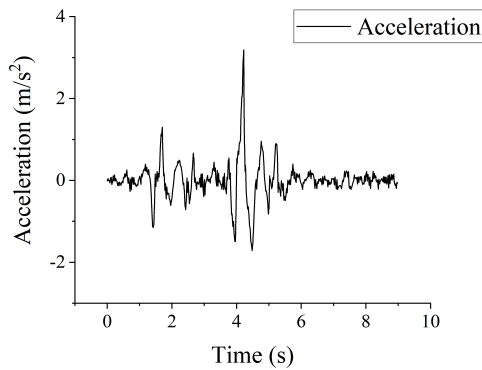


Fig. 15. Data after preprocessing (pit1—width of 47 cm and depth of 2.5 cm, and pit2—width of 35 cm and depth of 3.3 cm).

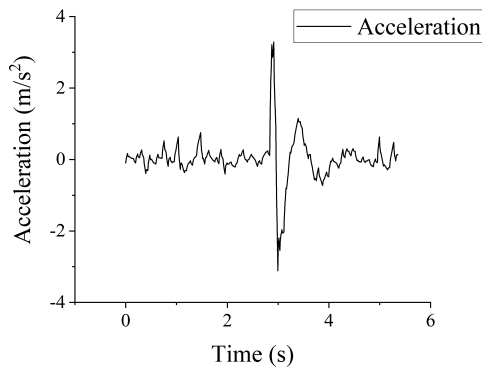


Fig. 16. Data after preprocessing (speed bump).

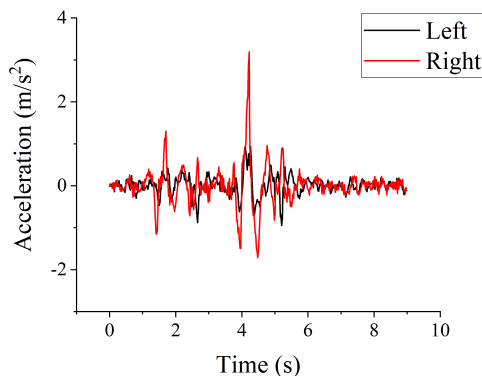


Fig. 17. Bilateral acceleration data.

with the largest absolute values, a set of acceleration data are integrated as shown in Fig. 18.

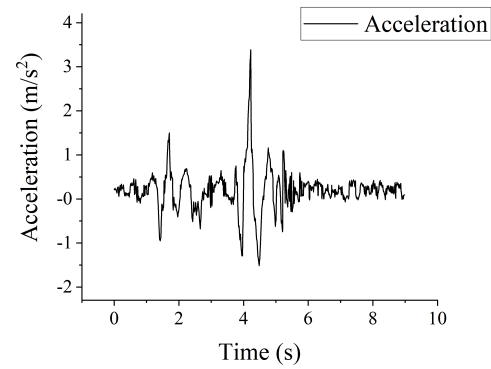


Fig. 18. Data fusion.

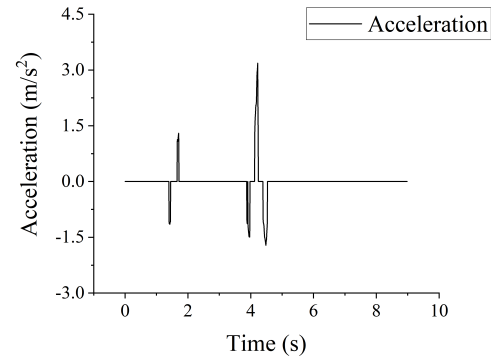


Fig. 19. Acceleration data denoising.

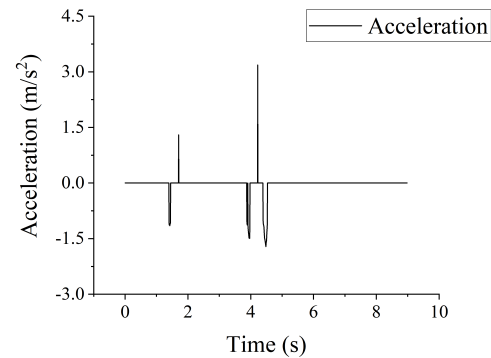


Fig. 20. Acceleration data peak normalization.

b) Noise filtering: When the car travels at different speeds, the peaks and troughs of acceleration caused by pits and speed bumps change significantly. However, the noise signal generated by road unevenness and body vibration also increases significantly. Therefore, based on the theoretical analysis in Section II and the large amount of data collected by the experiments, thresholds need to be set according to different speeds to filter out the noise signal and preserve the characteristic vibration signal, the filtered data are shown in Fig. 19.

c) Peak normalization: When a car hits a pit or a speed bump, the peak signal is often made up of many points. However, only one peak point of the wave peak should be maintained, and the other points should be deleted; otherwise, the same pit could be identified multiple times. After normalizing the data, the results are as shown in Fig. 20.

d) Identification of pits and speed bumps: Through the analysis of the obtained acceleration data, it is found that an obvious difference between the pits and the speed bumps during the

TABLE XVI

PIT AND SPEED BUMP IDENTIFICATION RESULTS

Speed	Pit number	Speed bump number	Pit identification	Speed bump identification
10~20 km/h	8	3	8	3
20~30 km/h	8	3	8	3
30~40 km/h	8	3	6	3

TABLE XVII

IDENTIFICATION RESULTS OF PITS AND SPEED BUMPS (10–20 cm)

Pit	Pit depth	Pit width	Detection depth	Detection width	Pass time
pit1	2.5 cm	47 cm	2~3 cm	more than 40 cm	3.47 s
pit2	3.3 cm	35 cm	3~5 cm	30~40 cm	15.76 s
pit3	2.8 cm	43 cm	2~3 cm	more than 40 cm	23.35 s
pit4	4.2 cm	27 cm	4~5 cm	20~30 cm	30.45 s
pit5	3.4 cm	83 cm	3~4 cm	more than 40 cm	53.14 s
pit6	4.3 cm	94 cm	4~5 cm	more than 40 cm	61.83 s
pit7	5.2 cm	65 cm	more than 50 cm	more than 40 cm	65.76 s
pit8	3.6 cm	27 cm	3~4 cm	30~40 cm	73.54 s

TABLE XVIII

LOCATION RECOGNITION RESULTS (10–20 cm)

Destination number	Distance	Destination number	Distance
pit1	12.5m	pit2	58.2m
pit3	87.6m	pit4	118.4m
pit5	199.3m	pit6	235.3m
pit7	252.1m	pit8	285.9m
speed bump1	165.7m	speed bump2	229.1m
speed bump3	269.6m		

driving process of the car is the sequence of the peaks and troughs. If there is a trough signal that meets the threshold requirements on the front side of the peak signal, it will be identified as a pothole; otherwise, it will be identified as a bump on the road. The identification results of pits and speed bumps are shown in Table XVI.

e) *Pit attribute identification:* When the car travels through different pits at the same speed, its peak acceleration will change significantly due to different depths of the pits. At the same time, GPS data are used to obtain the position information of the potholes and speed bumps, and the real speed and time difference between the peaks and troughs are used to identify the width information of the pit. The final recognition result is as shown in Table XVII and Table XVIII.

The test system can perfectly identify all the pits on the road section when the vehicle is running at a low speed. However, as the speed increases, the smaller pits will be submerged in the vibration and noise of the car body, resulting in data loss, but the accuracy is guaranteed to be 75%. The recognition of speed bump impact is excellent, and it can almost reach flawless recognition. To summarize, the proposed method has good measurement accuracy and meets pit and speed bump recognition requirements on low-grade roadways.

V. CONCLUSION

This article proposes a method for identifying road diseases and features using smartphone acceleration and GPS sensors. The collected sensor data are preprocessed to enhance accuracy by removing noise, and a genetic algorithm is applied to determine car model parameters. Based on the peak and trough data characteristics and the car model parameters, the EDFF technology is introduced to identify road surface pits and speed bumps. Simulations and experiments have confirmed the feasibility and effectiveness of the proposed approach. Compared with traditional detection methods, the proposed method significantly reduces the need for manpower, material resources, and costs, while meeting accuracy requirements. It is a practical and cost-effective solution for pit and speed bump recognition on low-grade roadways.

The future research focuses on improving detection accuracy, addressing missing data for small pits under high-speed conditions, and expanding the detection of various road conditions to establish a comprehensive database. In addition, utilizing mobile phones for the synchronized acquisition of GPS, acceleration, and other data, along with visual analysis of road condition information, is an area of interest to enhance the detection efficiency.

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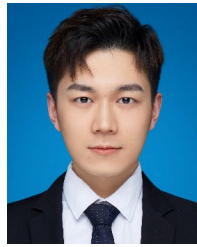
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Yunfei Yin (Member, IEEE) received the dual Ph.D. degrees in control science and engineering from the School of Astronautics, Harbin Institute of Technology (HIT), Harbin, China, and in automatic, electronic, and telecommunications engineering from the School of Engineering, Universidad de Sevilla, Seville, Spain, in 2021.

In 2021, he joined HIT, as an Assistant Professor, and as of 2024, he has advanced to the position of an Associate Professor. His research

interests include advanced control and artificial intelligence algorithms, and their applications on power electronic systems, intelligent connected vehicles, and intelligent transportation systems.



Wanli Fu received the master's degree in traffic and transportation engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2022.

His research interests include encompassing detection, evaluation, and maintenance techniques for pavement structure.



Xianyong Ma received the bachelor's degree in road, bridge, and river crossing engineering and the Ph.D. degree in highway and railway engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2014 and 2019, respectively.

From 2017 to 2018, he was a Visiting Ph.D. Student with the Department of Civil Engineering, Case Western Reserve University, Cleveland, OH, USA. He is currently a Postdoctoral Fellow and an Associate Professor in highway and railway engineering with HIT. His research interests include pavement mechanics, pavement health monitoring, fiber-optic sensors, and smart road.



Junpeng Yu received the bachelor's degree in traffic equipment and control engineering from the School of Traffic Science and Engineering, Harbin Institute of Technology, Harbin, China, in 2022, where he is currently pursuing the master's degree in transportation engineering, with a focus on image-based road crack detection.



Xinkai Li received the Ph.D. degree in engineering from the School of Transportation Science and Engineering, Harbin Institute of Technology, Harbin, China, in 2010.

In 2013, he assumed the position of an Associate Professor at the Harbin Institute of Technology. His research interests encompass the theoretical and methodological aspects of cement pavement structure design, as well as the detection, evaluation, and maintenance techniques for pavement structure and materials.



Zejiao Dong received the bachelor's and Ph.D. degrees in highway and railway engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2001 and 2006, respectively.

He joined as an Assistant Professor at HIT in 2004, where he was an Associate Professor in 2008. Since 2012, he has been working as a Full Professor with HIT. He was a Visiting Scholar with the University of Illinois at Urbana-Champaign, Champaign, IL, USA, from 2010 to 2011. He has authored or coauthored more than 70 articles in international and domestic journal articles and a book named *Dynamic Response of Asphalt Pavement* (China Science Press, 2015). His current research focuses on pavement mechanics, pavement health monitoring, and sustainable pavement materials.

Dr. Dong was a recipient of several prizes in the Provincial Science and Technology Progress Award. He served as the Co-Chair in Road Engineering for the World Transportation Conference.